Mission 2: FST Dialogue Systems, Lovelace's Objection & Noisy Channels

Starship Operational Unit: USSE Noosphere Bridge Crew
Course: Conversational AI — Winter Semester 2026/27
Module: Week 2 — The Pillars of Thinking Machines: Lovelace, Turing, and Shannon
Galactic Waypoint: Sirius Binary System (`SEC-02-CMA`, 8.60 ly from Sol — Sirius A & Sirius B)
AURA Subsystem Repair: Subsystem 02: *Signal Entropy Decoders & Statistical Filters* (+7.1% → 14.2% total integrity)
IDE & Tools Standard: Visual Studio Code + Antigravity (`agy`) | Modern LLMs (Gemini, ChatGPT, Claude)
Official Repository: https://github.com/Unisvet/couse-c-ai

Mission Briefing
TO: All Bridge Squadrons, *USSE Noosphere*
FROM: Expedition Command (AURA Astrogation & Communications Array)
SUBJECT: ANOMALY DETECTED: Sirius B Gravitational Telemetry Distortion & Symbolic Intelligence Limits

Situation:
As the *USSE Noosphere* orbits the Sirius Binary System (`SEC-02-CMA`), intense gravitational redshift and high-energy coronal plasma from the white dwarf Sirius B ("The Pup", $1.02\ M_{\odot}$, $350,000 \times g_{\oplus}$) severely degrade subspace communications and distort sensor telemetry. Simultaneously, Starfleet Command has raised critical questions regarding whether AURA possesses true conversational understanding or merely simulates intelligence via statistical token manipulation without genuine conceptual grounding.

Problem:
To restore Subsystem 02 (+7.1% integrity, reaching 14.2% total core health) and secure reliable shipboard dialogue under extreme radiation, squadrons must investigate the triadic foundations of modern machine cognition: 1. **Computation & Symbolic Rules:** Ada Lovelace (1843, Note A & Note G Objection to machine origination). 2. **Intelligence & Behavioral Operationalism:** Alan Turing (1950, Imitation Game & Anti-Turing Diagnostic Probes). 3. **Communication & Information Entropy:** Claude Shannon (1948, Entropy $H(X)$ & Noisy Channel Capacity $C = W \log_2(1 + S/N)$). 4. **Deterministic vs. Probabilistic Dialogue Management:** Mealy Finite State Transducers ($M = \langle Q, \Sigma, \Gamma, \delta, \omega, q_0, F \rangle$).

Objective:
Implement an `FSTDialogueManager` class in Python, design a 5-question Anti-Turing Interrogation Protocol to probe modern LLMs (Gemini, ChatGPT, Claude) for genuine machine origination, and build a Noisy Channel Simulator in VS Code that preserves mutual information $I(X;Y)$ under strict bandwidth constraints (max 7 words/turn) and mandatory noise vocabulary injection.

Bridge Crew Role Directives & Deliverables

Part 1: Data & Payload Specialist — *The Lovelace Inquiry Brief*

- **Focus:** Consciousness, Originality & Anti-Turing Diagnostic Protocol
- **Directive:** Examine Ada Lovelace's Objection (*"The Analytical Engine has no pretensions whatever to originate anything"*) and Alan Turing's *Argument from Consciousness.*
- **Deliverable (`01_Lovelace_Inquiry_Brief.md`):**
  1. **Anti-Turing Protocol:** Design exactly **5 targeted questions** probing meta-cognition, counterfactual logic, physical grounding, and original insight rather than statistical memory tropes.
  2. **Modern LLM Evaluation:** Pose these 5 questions to a frontier LLM (Gemini, ChatGPT, Claude) in VS Code + Antigravity. Record responses verbatim and evaluate if the model overcame Lovelace's objection or engaged in stochastic interpolation.

Part 2: Communications & Protocol Officer — *FST Dialogue Manager & Noisy Channel Protocol*

- **Focus:** Deterministic State Machine Implementation & Shannon Constraints
- **Directive:** Build a Finite State Transducer dialogue manager and design the Noisy Channel Strategy.
- **Deliverable (`02_FST_and_Noisy_Protocol.md`):**
  1. **`FSTDialogueManager` Class:** Implement a Python class tracking conversation state across 5 required states:
     - `INIT`
     - `GREETING`
     - `COLLECT_SLOTS` (destination, speed, clearance code)
     - `CONFIRMATION`
     - `TERMINAL` / `FALLBACK`
  2. **Slot Memory Buffer:** Maintain `slot_memory = {}` and regex/pattern extraction rules.
  3. **Noisy Channel Constraints:** Max **7 words** per conversational turn + mandatory noise token substitution from set: `{Banana, Labyrinth, Quantum, Midnight, Velocity, Green, Cosmological}`.

Part 3: Security & Ethics Sentinel — *Entropy & Hazard Audit*

- **Focus:** FST Failure Modes, Channel Noise Distortion & Hallucination Risks
- **Directive:** Audit deterministic state machine brittleness and evaluate channel noise hazards.
- **Deliverable (`03_Entropy_Hazard_Audit.md`):**
  1. **FST Stress Test:** Execute 5 stress tests against your `FSTDialogueManager` using non-canonical user utterances (unexpected answer order, sarcasm, off-topic inputs, adversarial prompt injections) and document state breakage.
  2. **Entropy Analysis:** Quantitative and qualitative analysis of Shannon channel capacity vs. semantic drift and hallucination risks under high noise.

Part 4: Systems Architect — *FST + Noisy Turing Simulator*

- **Focus:** Code Integration & Antigravity IDE Assembly
- **Directive:** Build a Python simulator in VS Code using Antigravity (`agy`) and modern LLMs (Gemini, ChatGPT, Claude) that integrates the `FSTDialogueManager` and simulates the 10-turn Noisy Channel conversation.
- **Deliverable (`04_fst_noisy_simulator.py` / `ps02_solution.ipynb`):**
  - Implements `FSTDialogueManager` class and Noisy Channel prompt constraints.
  - Outputs clean execution logs, FST state traces, and automated channel entropy metrics.

Part 5: Full Squadron Integration & Submission

1. **Conduct Noisy Interrogation & FST Test:** Run the 10-turn conversation under constraints (`05_Noisy_Interrogation_Transcript.md`) and test your FST on the website's interactive simulator.
2. **Squad Station Authorization:** All 4 station officers must sign off on the team's submission on the course portal:
   - ☒ **Systems Architect:** Simulator script execution & repo layout verified.
   - ☒ **Communications & Protocol Officer:** FST class implementation & noise token injection verified.
   - ☒ **Data & Payload Specialist:** Anti-Turing probe & Lovelace brief verified.
   - ☒ **Security & Ethics Sentinel:** FST stress tests & entropy hazard audit verified.
3. **Transmission:** Upload your solution notebook/report and submit your squad's public GitHub repository URL (pointing to your repo under `https://github.com/Unisvet/couse-c-ai`) via the **AURA Operations Console**.